

Domácí úkol č. 1

1.1

Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle \cdot, \cdot \rangle$ platí pro libovolné $u, v \in V$:

$$\begin{aligned}\operatorname{Re}(\langle u, v \rangle) &= \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2), \\ \operatorname{Im}(\langle u, v \rangle) &= \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2).\end{aligned}$$

Důkaz pro reálnou část:

$$\begin{aligned}\operatorname{Re}(\langle u, v \rangle) &= \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2) \\ &\stackrel{(\text{Pomocí SSS SL2})}{=} \frac{1}{4}(\langle u, u \rangle + \langle u, v \rangle + \overline{\langle u, v \rangle} + \langle v, v \rangle - (\langle u, u \rangle - \langle u, v \rangle - \overline{\langle u, v \rangle} + \langle v, v \rangle)) \\ &= \frac{1}{4}(\|u\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} + \|v\|^2 - \|u\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} - \|v\|^2) \\ &= \frac{1}{4}(2\langle u, v \rangle + 2\overline{\langle u, v \rangle}) \\ &\stackrel{x + \bar{x} = 2\operatorname{Re}(x)}{\stackrel{\downarrow}{=}} \frac{1}{4}(2 \cdot 2\operatorname{Re}(\langle u, v \rangle)) = \operatorname{Re}(\langle u, v \rangle)\end{aligned}$$

Důkaz pro imaginární část:

$$\begin{aligned}\operatorname{Im}(\langle u, v \rangle) &= \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2) \\ &\stackrel{(\text{Pomocí SSS SL2})}{=} \frac{1}{4}(\langle u, u \rangle + \langle u, -iv \rangle + \overline{\langle u, -iv \rangle} + \langle -iv, -iv \rangle - (\langle u, u \rangle + \langle u, iv \rangle + \overline{\langle u, iv \rangle} + \langle iv, iv \rangle)) \\ &= \frac{1}{4}(\|u\|^2 + i\langle u, v \rangle - i\overline{\langle u, v \rangle} + \|v\|^2 - (\|u\|^2 - i\langle u, v \rangle + i\overline{\langle u, v \rangle} + \|v\|^2)) \\ &= \frac{1}{4}(2i\langle u, v \rangle - 2i\overline{\langle u, v \rangle}) = \frac{1}{4}(-2i(\langle u, v \rangle - \overline{\langle u, v \rangle})) \\ &\stackrel{-ix + i\bar{x} = 2\operatorname{Im}(x)}{\stackrel{\downarrow}{=}} \frac{1}{4}(2 \cdot 2\operatorname{Im}(\langle u, v \rangle)) = \operatorname{Im}(\langle u, v \rangle)\end{aligned}$$

Poznámka k identitám:

Uvažujme vektor $x = (a_1 + b_1i, a_2 + b_2i, \dots, a_n + b_ni)^T$ nad komplexním vektorovým prostorem $\mathbb{C}^n$, kde $n \in \mathbb{N}$. Pak platí:

$$\bullet \quad x + \bar{x} = \begin{pmatrix} a_1 + b_1i \\ \vdots \\ a_n + b_ni \end{pmatrix} + \begin{pmatrix} a_1 - b_1i \\ \vdots \\ a_n - b_ni \end{pmatrix} = \begin{pmatrix} 2a_1 \\ \vdots \\ 2a_n \end{pmatrix} = 2 \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = 2\operatorname{Re}(x)$$

$$\bullet \quad -ix + i\bar{x} = -i(x - \bar{x}) = -i \begin{pmatrix} 2b_1 i \\ \vdots \\ 2b_n i \end{pmatrix} = -2i^2 \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 2 \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 2\text{Im}(x)$$